

YASHI SHRIVASTAVA

Staff AI Systems Engineer & LLM Infrastructure Architect

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PROFESSIONAL SUMMARY

Staff AI & Machine Learning Systems Engineer with 8+ years of experience designing, scaling, and deploying frontier generative AI, distributed LLM inference, autonomous agentic systems, and high-throughput ML infrastructure across top-tier Silicon Valley tech companies (Databricks, Scale AI, Uber AI Labs). Deep expertise in LLM fine-tuning (LoRA/QLoRA, DPO/RLHF), agentic orchestration frameworks, low-latency hybrid RAG pipelines, model quantization (AWQ/GPTQ), and speculative decoding on large GPU clusters. Proven track record of reducing p99 inference latency by 42%, scaling vector search across 50M+ embeddings, and orchestrating production multi-agent architectures serving tens of millions of daily queries.

CORE TECHNICAL CAPABILITIES

GenAI & Agentic Architectures: Agentic reasoning (ReAct, Plan-and-Solve, Multi-Agent Swarms), tool use & function calling, structured generation, self-correcting validation loops, prompt engineering & evals (LLM-as-a-judge, MT-Bench), guardrails & alignment (RLHF, DPO, constitutional AI).

LLM Systems & Inference Optimization: vLLM, TensorRT-LLM, Triton Inference Server, speculative decoding, continuous batching, PagedAttention, KV-cache quantization (FP8, INT4, AWQ), model parallelism (Tensor/Pipeline/Expert Parallelism), Ray Serve.

Distributed Training & Fine-Tuning: PyTorch, DeepSpeed (ZeRO 1/2/3), Megatron-LM, Hugging Face (Transformers, PEFT, TRL), Ray Train, LoRA/QLoRA, full parameter fine-tuning, SLURM, Kubernetes (KubeFlow, Volcano).

Information Retrieval & RAG: Hybrid dense/sparse retrieval (BM25 + SPLADE + dense embeddings), ColBERT token-level re-ranking, vector databases (Qdrant, Pinecone, Milvus, pgvector), chunking strategies, semantic routing, context compression.

Platforms & Languages: Python, C++, Rust, SQL, TypeScript/Node.js, CUDA, Linux, FastAPI, Docker, Kubernetes, AWS (SageMaker, EKS, EC2 P4d/P5), GCP (Vertex AI, Cloud TPU/GPU), Azure AI, Prometheus, Grafana, MLflow, Weights & Biases.

PROFESSIONAL EXPERIENCE

Databricks — Staff AI Engineer, Generative AI & Foundation Models Platform

San Francisco, CA | Apr 2023 – Present

- Architected core runtime for enterprise LLM serving platform powering managed endpoints for 1,200+ enterprise customers, achieving 42% p99 latency reduction and \$3.2M annual GPU cost savings.
- Designed and implemented an autonomous multi-agent tool execution engine with dynamic backtracking, sandboxed Python code execution, and deterministic validation schema enforcement.
- Built distributed speculative decoding pipelines pairing compact draft models with 70B+ parameter foundation models, boosting token generation throughput by 2.4x.
- Led end-to-end evaluation harness and safety guardrail infrastructure, automating red-teaming, jailbreak detection, and hallucination scoring across customer-facing agent deployments.
- Mentored 9 machine learning engineers and served on the AI Architecture Review Board, establishing best practices for model quantization, inference caching, and latency budgets.

Scale AI — Senior AI Systems Engineer, Enterprise GenAI & Frontier Alignment

San Francisco, CA | Aug 2021 – Mar 2023

- Built scalable alignment pipelines for frontier LLMs, implementing RLHF and Direct Preference Optimization (DPO) pipelines handling over 5M pairwise reward ratings.
- Architected production enterprise RAG systems across Fortune 500 financial and legal datasets, combining Qdrant vector indexing, BM25 keyword matching, and cross-encoder re-ranking (89.4% precision).
- Engineered continuous model quantization pipelines (AWQ, GPTQ, INT8 weight-only) to compress 13B and 70B parameter models onto single-GPU instances without perceptual degradation.
- Developed real-time telemetry and streaming evaluation pipelines detecting prompt injection, data leakage, and drift across live client inference sessions.

Uber AI Labs / Michelangelo Platform — Machine Learning Engineer → Senior ML Engineer

San Francisco, CA / Seattle, WA | Jun 2018 – Jul 2021

- Engineered high-throughput, low-latency deep learning inference pipelines on the Michelangelo ML platform, supporting 500k+ real-time QPS for dynamic dispatch and ETA prediction.
- Migrated legacy deep learning training workflows to distributed PyTorch on Kubernetes with Horovod and DeepSpeed, cutting multi-GPU training time from 72 hours to 18 hours.
- Built an automated model monitoring and feature drift detection service tracking 200+ production models, slashing silent performance degradation incidents by 65%.
- Co-developed a unified feature store caching real-time geospatial and user features with sub-5ms read latency.

SELECTED PROJECTS

AegisAgent — Distributed Agentic Orchestration & Tool-Execution Engine

Technologies: Python, FastAPI, Ray, LangGraph, vLLM, Redis, Docker, OpenTelemetry

Asynchronous, high-resilience agent runtime supporting DAG-based multi-step tool execution, state checkpointing, and dynamic backtracking. Features human-in-the-loop verification, isolated code execution sandboxes, and parallel tool calling. Deployed across enterprise suites handling 15M+ monthly tool executions (99.96% reliability).

OmniRAG — Low-Latency Enterprise Hybrid Retrieval Engine

Technologies: PyTorch, Qdrant, FAISS, pgvector, Hugging Face Transformers, Ray Serve, FastAPI

Modular retrieval-augmented generation engine combining dense-sparse vector indexing (SPLADE + BGE-large), ColBERT token-level re-ranking, and dynamic context compression with semantic caching. Achieved sub-45ms p95 latency across 50M+ documents with an 89.4% factual accuracy score.

SpecInfer-Dist — High-Throughput Distributed Speculative Decoding System

Technologies: CUDA, C++, PyTorch, TensorRT-LLM, vLLM, Triton Inference Server

Open-source distributed inference framework pairing small draft models with 70B+ parameter target LLMs across multi-node clusters. Yielded a 2.4x wall-clock speedup for code generation and structured JSON extraction without quality degradation.

EDUCATION

Carnegie Mellon University (CMU) — Master of Science (M.S.) in Computer Science (AI & ML) | 2016 – 2018 | GPA: 3.92/4.0

Indian Institute of Technology (IIT), Delhi — Bachelor of Technology (B.Tech) in Computer Science & Engineering | 2012 – 2016 | High Distinction

PUBLICATIONS, HONORS & LANGUAGES

- **Publications & Talks:** ICML 2024 Workshop (Speculative Decoding for JSON at Scale); Keynote at AI Engineer World's Fair SF (2024); EMNLP 2023 (Enterprise Hybrid RAG).
- **Certifications & Honors:** AWS Certified Machine Learning - Specialty; NVIDIA Certified Associate - GenAI; Databricks Global Hackathon Winner (GenAI Track).
- **Languages:** English (Full professional proficiency), Hindi (Native proficiency).